

Auteur: Stan Van Campenhout

Studierichting: Informatica

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$$\begin{aligned} & \sum_{n=2}^{\infty} n^2 e^{-n^3} \\ & \int_0^{+\infty} n^2 e^{-n^3} dn = -\frac{1}{3} \int_0^{+\infty} e^{-n^3} d(-n^3) \\ & = -\frac{1}{3} \left[ e^{-n^3} \right]_0^{+\infty} = -\frac{1}{3} \left( \lim_{a \rightarrow +\infty} \frac{1}{e^{a^3}} - \frac{1}{e^{0^3}} \right) \\ & = -\frac{1}{3} (0 - 1) = \frac{1}{3} \Rightarrow \text{bestaat} \\ & \text{integraalcriterium} \Rightarrow \text{convergent} \end{aligned}$$

## References